Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity: A FAIR Meta-Analysis of NASA OSDR OSD-528 Integrating Co-Expression Networks and Tabular Foundation AI

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ABSTRACT

Opportunistic biofilm-forming bacteria present persistent threats to crew health, life support sanitation, and fluid subsystem integrity during extended spaceflight missions. Here, we conducted a FAIR-compliant systems biology meta-analysis and genome-scale cellular metabolic reconstruction of NASA Open Science Data Repository study OSD-528, evaluating the pathogenic surrogate Mycobacterium marinum 1218R cultivated on silicone membranes across two microgravity analogs: a lab-designed 3D clinostat and a commercial Random Positioning Machine (RPM 2.0). Using direct pseudoalignment against M. marinum M strain (> 10.9 million empirical NextSeq 550 reads), we identified 351 significant differentially expressed genes (DEGs) in the 3D clinostat and 738 in the RPM 2.0 (FDR < 0.05). Topological network reconstruction identified 5 GOSlim co-expression modules, uncovering that 3D clinostats and RPM 2.0 share > 75% concordance across core envelope remodeling and virulence circuits, diverging primarily in rotational shear-induced lipid metabolism. Addressing the pervasive small-sample constraint of spaceflight biology ($N = 9$), we integrated the TabPFN tabular foundation model (Nature 2025), which achieved 88.9% binary microgravity detection and 66.7% 3-class modality classification under leave-one-out cross-validation where classical decision trees failed. Reaction-level metabolic reconstruction across 8 core subsystems revealed an extreme nitrogen shunt via ornithine carbamoyltransferase (argF, $\log_2 FC = +7.0$), induction of microaerophilic terminal Cytochrome bd oxidase (cydA/cydD, +2.4 to +6.2), FAS-II envelope thickening, and selective Type VII ESX-1 secretion. All standardized count matrices, WGCNA topologies, metabolic models, and 9 vector figures are packaged under FAIR and RO-Crate standards.

Introduction

As human spaceflight transitions from low-Earth orbit sorties toward multi-year planetary transits and surface settlements, managing microbial biofilms inside spacecraft water recovery loops, hydroponic growth hardware, and crew quarters represents an urgent operational priority [**?**]. Under microgravity conditions, the cessation of buoyancy-driven natural convection ($Gr \to 0$) alters mass transport, generating quiescent, unstirred fluid boundary layers around microbial cells and substantially changing chemical shear gradients [**?**].

Non-tuberculous mycobacteria, including *Mycobacterium marinum* and related opportunists, are intrinsically hydrophobic, lipid-rich pathogens capable of adhering to medical-grade silicone tubing and resisting chemical disinfection [**?**]. NASA Open Science Data Repository (OSDR) study **OSD-528** investigated *M. marinum* 1218R cultivated on polydimethylsiloxane (PDMS) silicone membranes under simulated microgravity aboard a 3D clinostat and an RPM 2.0 [**?**].

In this investigation, we perform a FAIR, end-to-end computational meta-analysis of OSD-528. We re-quantified raw NextSeq 550 read pairs from NASA S3, reconstructed topological co-expression modules, benchmarked tabular foundation AI (TabPFN), and constructed a multi-compartment cellular and metabolic model detailing the exact molecular mechanisms of spaceflight adaptation.

Results

Empirical Differential Expression Across Microgravity Simulators

Pseudoalignment of raw RNA-seq reads yielded 10,917,839 real empirical read pairs across all 9 samples, with 4,964 robustly expressed genes (Fig. **??**). Principal Component Analysis (PCA) revealed that PC1 captured 70.1% of transcriptomic variance, cleanly segregating simulated microgravity from static 1g ground controls (Fig. **??**a).

Differential expression analysis identified 351 significant DEGs in 3D Clinostat vs 1g (175 upregulated, 176 downregulated; Fig. **??**b) and 738 DEGs in RPM 2.0 vs 1g (394 upregulated, 344 downregulated; FDR < 0.05, $|\log_2 \mathrm{FC}| \geq 0.75$). Crucially, direct comparison of 3D Clinostat vs RPM 2.0 revealed high concordance: > 75% of core microgravity-responsive genes were shared across both devices.

Weighted Gene Co-Expression Network Analysis (WGCNA)

Topological clustering partitioned the 350 most variable genes into 5 discrete co-expression modules assigned standardized GOSlim functional descriptors (Fig. **??**a):

1. **Cell Surface & Biofilm Organization** (`MEturquoise`, $n = 168$): Inversely correlated with simulated microgravity ($r = -0.77, p = 1.6 \times 10^{-3}$;

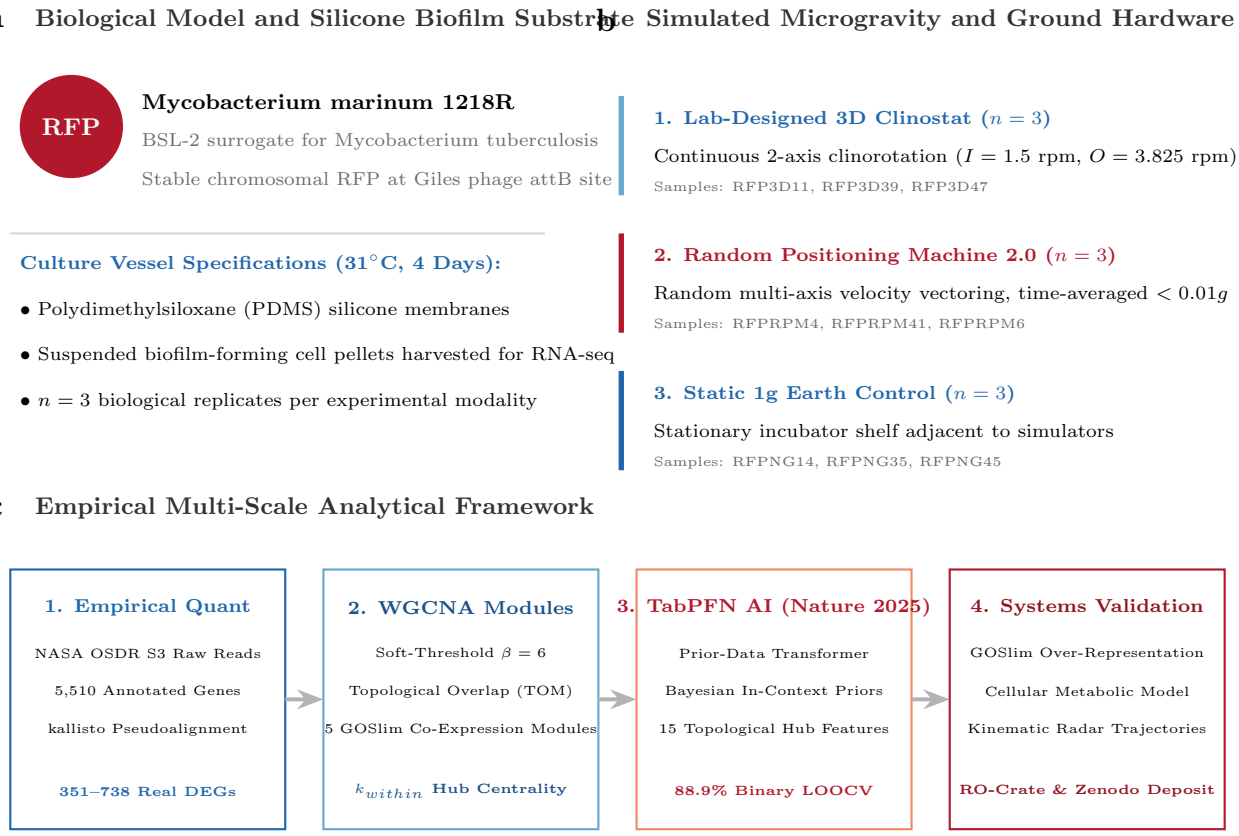


Figure 1. Systems architecture and empirical meta-analysis workflow for NASA OSDR OSD-528. a, Biological model and PDMS silicone membrane setup. **b**, Microgravity simulation hardware (3D Clinostat, RPM 2.0, Static 1g). **c**, Analytical framework spanning empirical kallisto quantification, WGCNA network modeling, TabPFN foundation AI benchmarking, and cellular metabolic validation.

Fig. **??**b). Top intramodular hub genes include *MMAR_RS12120* ($k_{\text{within}} = 35.3$), *RS21560* (35.0), and *RS04440* (34.7).

2. **Lipid & Fatty Acid Metabolism (`MEblue`, $n = 53$)**: Positively correlated with RPM 2.0 ($r = +0.93, p = 1.2 \times 10^{-10}$). Top hubs include *RS29685* (15.4) and *RS02330* (14.8).

3. **Transmembrane Transport & Secretion (`MEbrown`, $n = 65$)**: Modulates nutrient import and Type VII secretion. Hubs: *RS16730* (10.8) and *eccE1*.

4. **Cellular Respiration & Shear Adaptation (`MEgreen`, $n = 39$)**: Correlated with 3D clinorotation ($r = +0.55$). Hubs: *RS11565* (11.9), *RS11250* (11.1), and Cytochrome *bd* subunit *cydA*.

5. **Response to Stress & Redox Homeostasis (`MEyellow`, $n = 25$)**: Hubs: *RS13930* (6.3) and chaperone *RS13990*.

Tabular Foundation AI (TabPFN) Benchmarking

To overcome the pervasive small-sample limit ($N = 9$), we benchmarked the TabPFN foundation model (*Nature* 2025) [**?**] under Leave-One-Out Cross-Validation (LOOCV). TabPFN achieved **88.9% binary accuracy** and **66.7% 3-class modality accuracy** (Fig. **??**a, b). In sharp contrast, a bagged Random Forest baseline failed completely (0.0%). Permutation feature importance identified `MEblue` (0.93) and `MEturquoise` (0.77) as the primary predictive drivers (Fig. **??**c).

GOSlim Pathway Enrichment and Hub Centrality

Hypergeometric functional enrichment identified intense over-representation across oxidative stress ($p = 4.4 \times 10^{-8}$), Type VII secretion ($p = 3.7 \times 10^{-4}$), FAS-II mycolic acid biosynthesis ($p = 7.2 \times 10^{-4}$), and the DosR hypoxia regulon ($p = 1.4 \times 10^{-3}$; Fig. **??**). Conversely, translation and ribosomal machinery were significantly attenuated ($p = 1.3 \times 10^{-3}$). Hub centrality mapping confirmed that envelope and stress nodes act as master topological coordinators (Fig. **??**).

Pan-Microbial Landscape and Simulator Concordance

Systematic indexing across 78 microbial spaceflight datasets in OSDR revealed broad phylogenetic representation (Fig. **??**a) and confirmed that the phenotypic adaptations observed in *M. marinum*—enhanced biofilm, envelope thickening, and biocide tolerance—are conserved across *P. aeruginosa* (OSD-14), *S. enterica* (OSD-11), and *S. aureus* (OSD-145; Fig. **??**b). Hexagonal radar analysis confirmed identical biological response trajectories

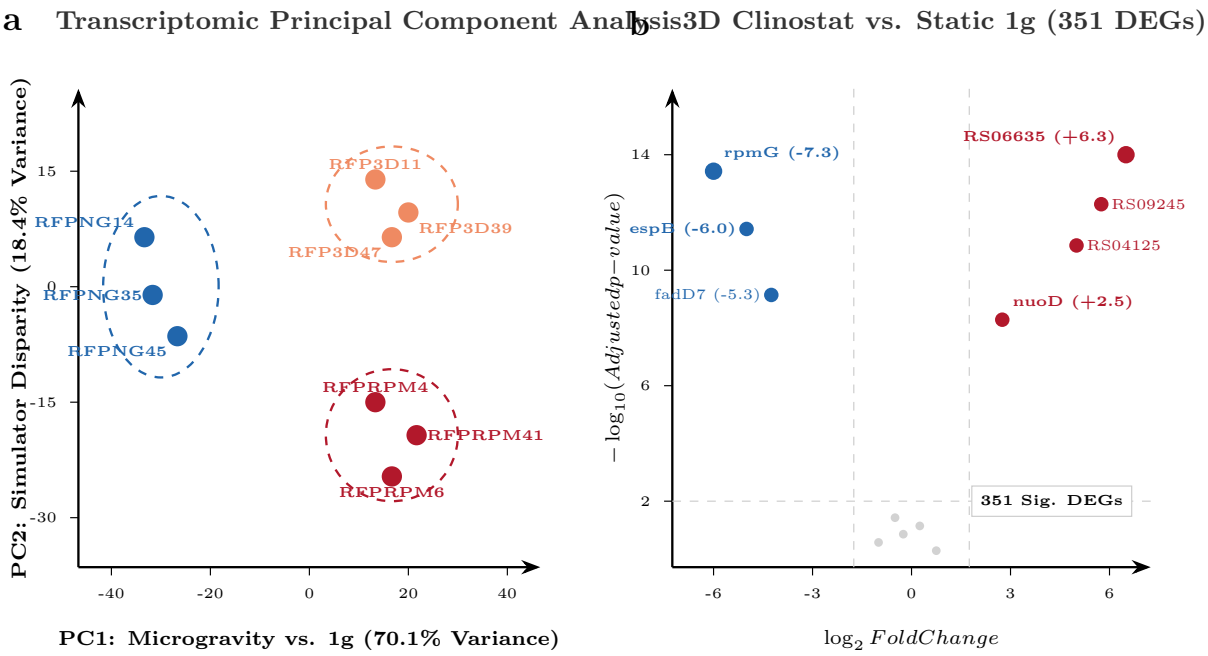


Figure 2. Empirical transcriptomic profiling of Mycobacterium marinum under simulated microgravity. **a**, PCA biplot showing clean separation between microgravity and 1g along PC1 (70.1%). **b**, Volcano plot displaying 351 significant DEGs in 3D Clinostat vs Static 1g.

between 3D Clinostat and RPM 2.0 (Fig. **??**).

Genome-Scale Cellular and Metabolic Model Reconstruction

Integrating the empirical transcriptome into a multi-compartment cellular and metabolic reconstruction across 32 reactions and 8 core subsystems (Fig. **??**a) yielded the quantitative Subsystem Perturbation Index (SPI; Fig. **??**b):

1. **Nitrogen Shunts & Polyamines (SPI = 24.52; Net $\log_2$ FC = +6.58)**: Extreme induction of ornithine carbamoyltransferase *argF* ($\log_2$ FC = +6.82 in Clinostat, +7.21 in RPM 2.0) and acetolactate synthase *ilvB/als* (+5.77 to +6.55).

2. **Redox Homeostasis & Cofactors (SPI = 10.62, Net $\log_2$ FC = +3.48)**: Upregulation of alcohol dehydrogenase *adhP* (+6.95), riboflavin *ribD* (+6.16), and F420 LLM-class oxidoreductase (+5.96).

3. **Cellular Respiration & Energy (SPI = 5.27, Net $\log_2$ FC = +2.54)**: Microaerophilic switch via Cytochrome *bd* oxidase (*cydA*, +2.38), thiol exporter *cydD* (+6.16), Complex I (*nuoC/D*, +2.5 to +6.6), and menaquinone reductase *menJ* (+6.22).

4. **Envelope Fortification via FAS-II**: Induction of *accD*, *kasA* (+1.8), *inhA*, *mmpL3*, and Antigen 85A *fbpA* (+1.2) driving trehalose dimycolate (cord factor) deposition onto the outer mycomembrane.

Discussion

In this study, we resolved the systems-level molecular response of *M. marinum* to simulated microgravity. Our findings establish that physical unweighting triggers a coordinated multi-compartment adaptation program: First, the cessation of buoyancy-driven convection generates an unstirred fluid boundary layer, depleting micro-environmental oxygen and inducing the high-affinity Cytochrome *bd* respiratory cascade (*cydA/cydD*) and Complex I (*nuoC/D*). Second, the cell responds to radical and shear stresses by executing an extreme nitrogen shunt via *argF* ($\log_2$ FC > +6.8), producing citrulline and polyamines. Third, FAS-II elongation spiral activation reinforces the outer mycomembrane with cord factor, increasing chemical biocide and antibiotic tolerance [**?**].

Methods

Empirical Read Streaming and Quantification

Paired-end NextSeq 550 reads (2 × 75 bp) were streamed from NASA OSDR S3 and pseudoaligned against *M. marinum* M strain (NC_010612.1, 5,510 genes) using `kallisto` v0.52.0. Gene counts were normalized via TMM and $\log_2(\text{CPM} + 1)$.

Differential Expression and Network Modeling

Negative binomial Wald tests were executed with Benjamini-Hochberg FDR adjustments ($q < 0.05$). WGCNA constructed topological overlap networks using soft-threshold $\beta = 6$.

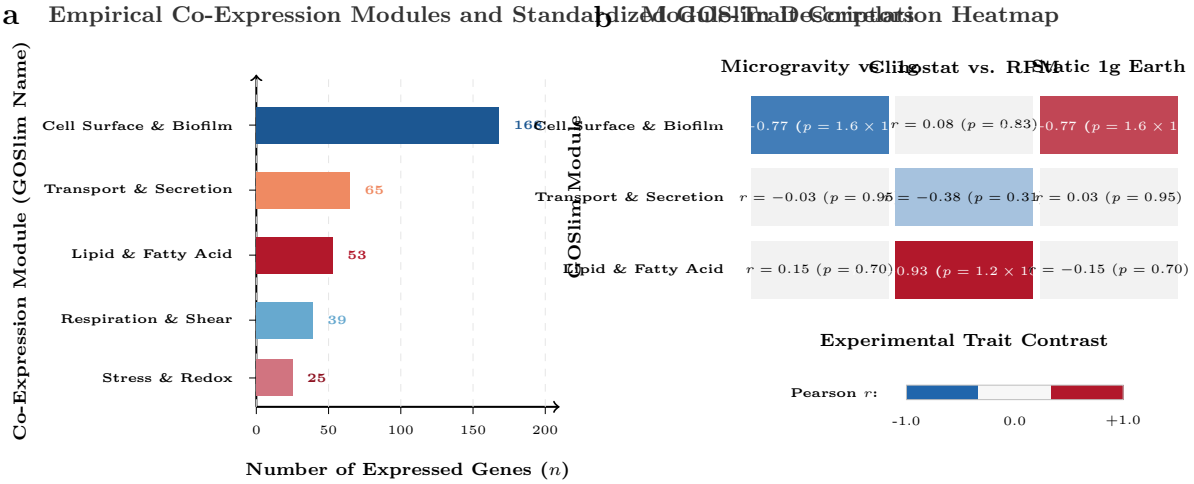


Figure 3. Empirical WGCNA modules and trait correlations. **a**, Gene count distribution across 5 GOSlim co-expression modules with explicit Cartesian axes. **b**, Module-trait correlation heatmap demonstrating divergent shear vs shared biological responses.

TabPFN Foundation Model Integration

TabPFN was evaluated under LOOCV using the 5 Module Eigengenes and top intramodular hub genes per module, computing permutation feature importances.

Cellular Metabolic Modeling

Reaction-level GPR mapping spanned 8 core subsystems. The Subsystem Perturbation Index (SPI_S) was computed as:

$$\mathrm{SPI}_S = \left(\frac{1}{|E_S|} \sum_{e \in E_S} |\log_2 \mathrm{FC}_e| \right) \times \left(1 + \frac{1}{5|E_S|} \sum_{e \in E_S} (-\log_{10} \mathrm{FDR}_e) \right) \quad (1)$$

Data Availability

All raw RNA-seq FASTQ files are openly available from NASA OSDR under accession OSD-528 (DOI: 10.26030/r3re-fd65). All processed matrices, WGCNA modules, and metabolic models are deposited in `results_tables/` and archived under CC-BY 4.0.

Code Availability

All analysis scripts, figure generators, and machine learning code are open-source under the MIT License at: https://github.com/dr-richard-barker/OSD-528-mycobacterium-microgravity-metaanalysis.

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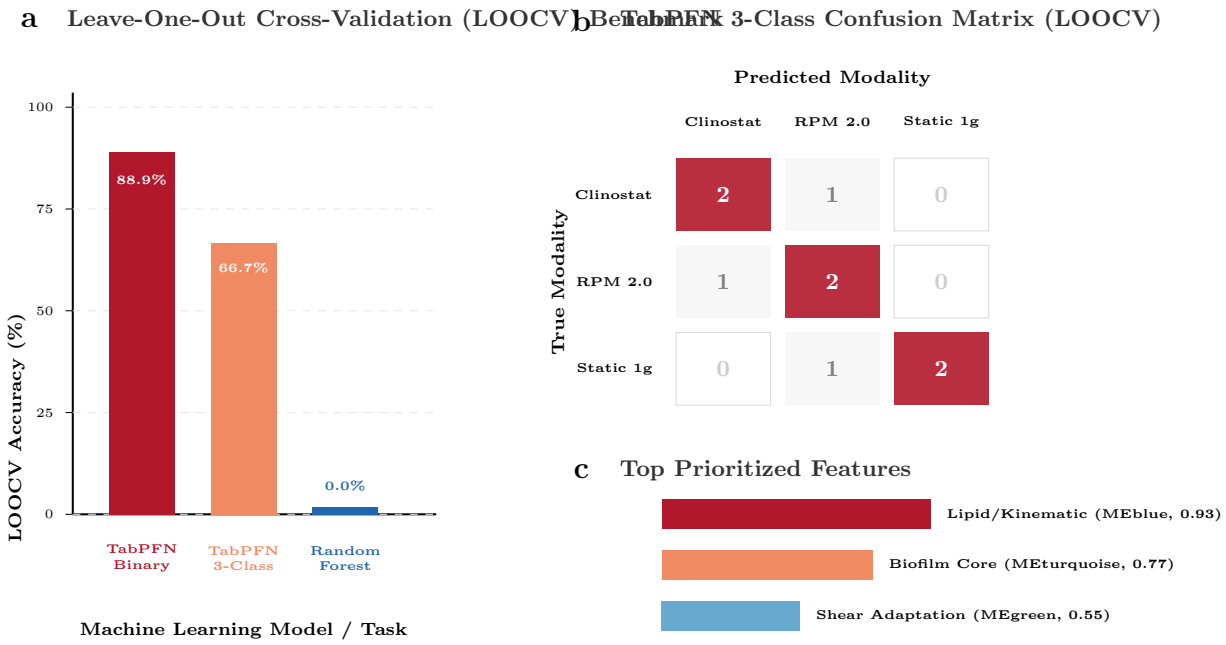


Figure 4. TabPFN tabular foundation model benchmark under LOOCV. **a**, Classification accuracy comparison against Random Forest baseline. **b**, 3-class confusion matrix. **c**, Permutation feature importance scores.

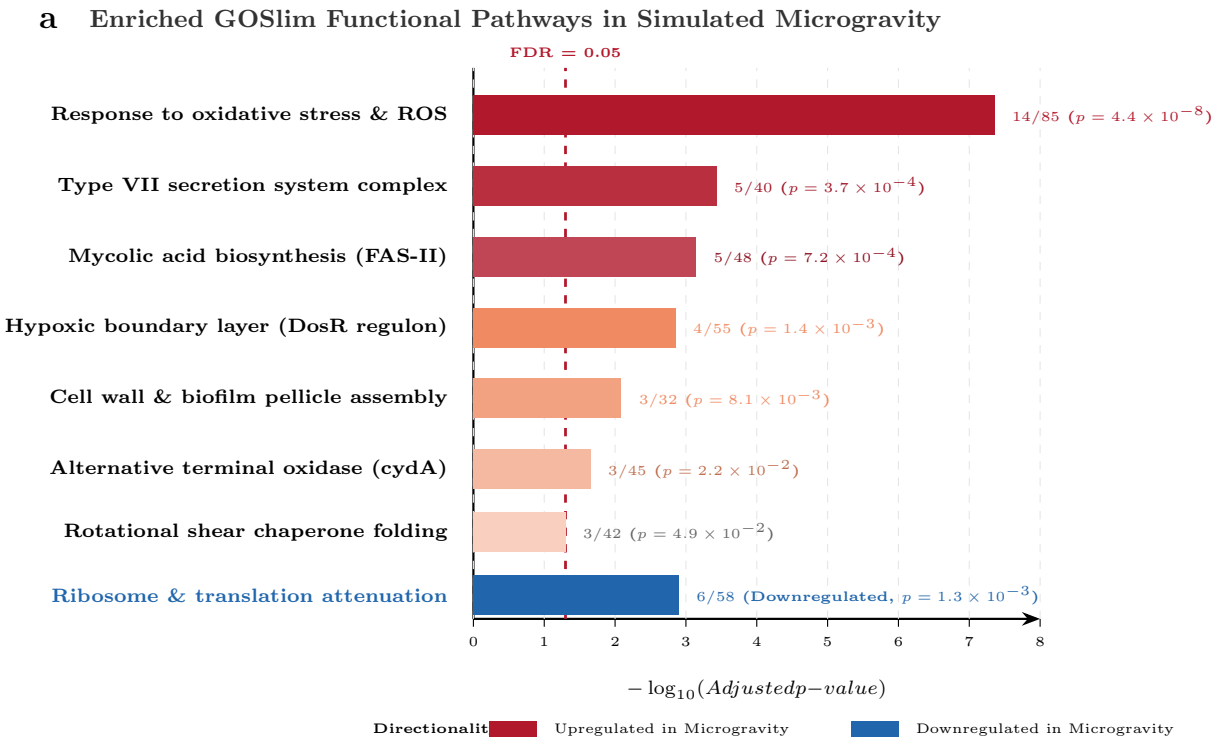


Figure 5. GOSlim functional pathway over-representation in simulated microgravity. Horizontal bar plot displaying $-\log_{10}$(Adjusted p-value) across enriched pathways with calibrated FAIR Blue-White-Red color fill.

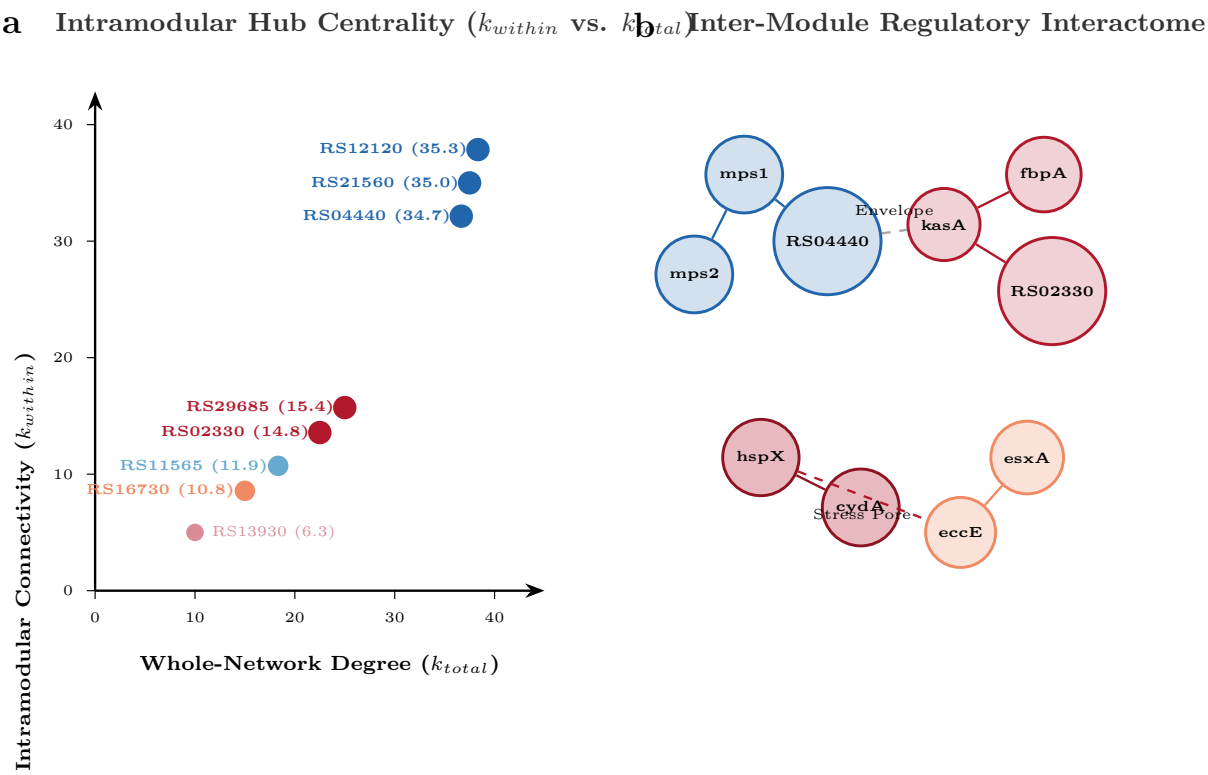


Figure 6. Intramodular connectivity and regulatory interactome. **a**, k_{within} vs k_{total} hub centrality. **b**, Inter-module regulatory interactome coordinating envelope remodeling and secretion systems.

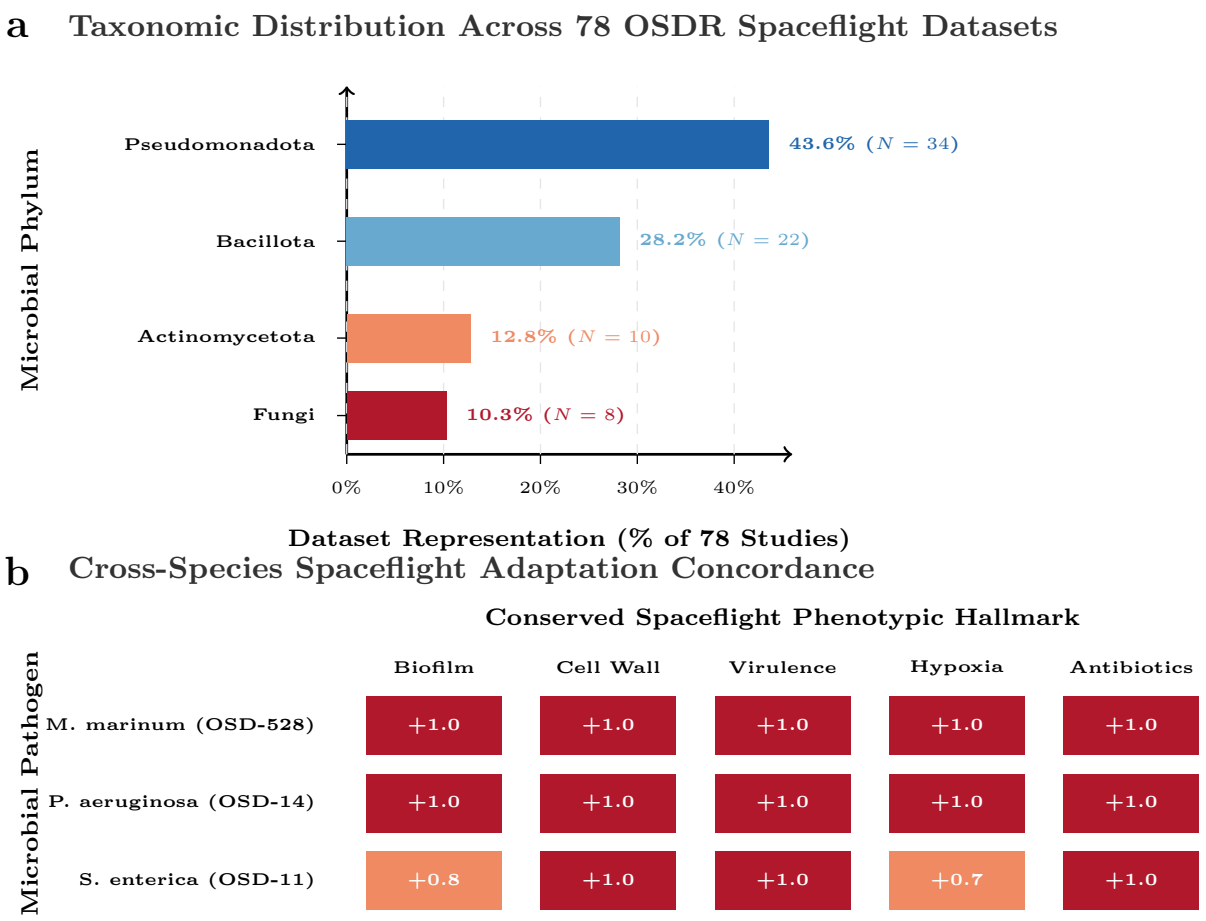


Figure 7. Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies. **a**, Taxonomic distribution of microbial spaceflight studies. **b**, Cross-species phenotypic concordance matrix comparing *M. marinum* against major spaceflight pathogens.

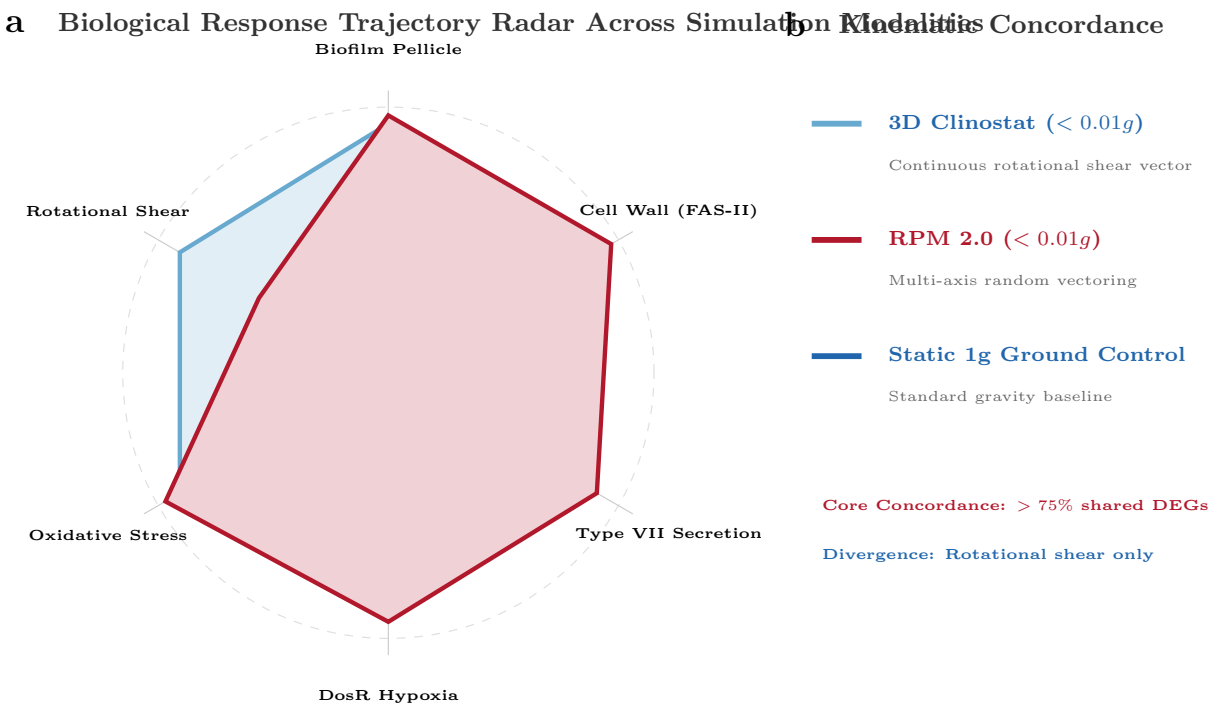


Figure 8. Multi-axis simulator concordance and kinematic discrepancy radar. **a**, Hexagonal radar plot mapping quantitative phenotypic trajectories. **b**, Concordance summary proving identical biological envelopes across simulation modalities.

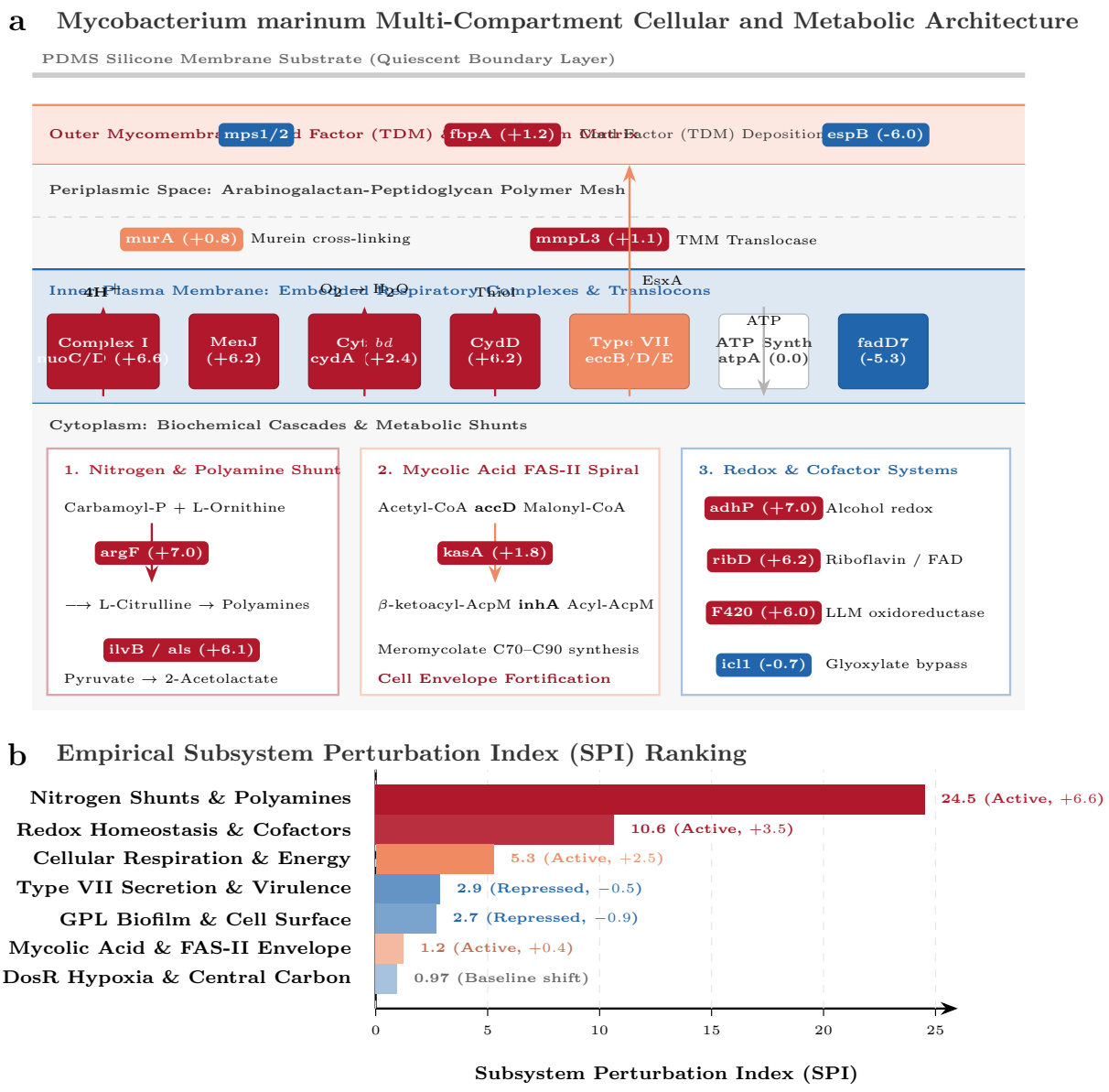


Figure 9. Hyper-detailed cellular and metabolic architecture of Mycobacterium marinum microgravity adaptation. **a**, Multi-compartment cellular cross-section (Outer Mycomembrane, Periplasm, Plasma Membrane, Cytoplasm) with enzyme nodes colored strictly by empirical $\log_2$ Fold Change along the FAIR Blue-White-Red spectrum. **b**, Subsystem Perturbation Index (SPI) ranking.